

Flight Phase: Cruise

Input name	Value	Label	Degree
Turbulence	2.32	Moderate	0.28
Air speed	219.93	High	0.34
Altitude	11403.69	Medium	0.64
Pitch angle	-8.31	Nose down	0.33
Vertical speed	619.86	Level	0.38
Engine health	1.04	Degraded	0.99
Fuel status	96.11	Full	1.00
Proximity to obstacles	251.76	Near	0.65
TCA's conflict level	0.32	None	0.57

Output name	Value	Label	Degree
Thrust command	77.50	High	0.83
Elevator deflection	0.00	Neutral	1.00
Autopilot mode suggestion	1.50	Approach	0.25
Stall warning	1.00	Advisory	1.00
Fuel reserve command	1.00	Economy mode	1.00
Stress reduction command	1.00	Increase autopilot	1.00

Explanation
- Low airspeed and pitch attitude indicate high stall risk; issuing stall advisory is critical. - Engine health degraded; thrust adjusted to ensure performance under failure conditions.

Main Contributing Factors		
Engine health	1.04	degraded
Fuel status	96.11	full

Flight Phase: Approach

Input name	Value	Label	Degree
Turbulence	2.86	Severe	0.78
Air speed	103.27	Slow	0.66
Altitude	756.39	Low	1.00
Pitch angle	16.93	Nose up	0.61
Vertical speed	1714.36	Climb	0.79
Engine health	0.69	Degraded	0.59
Fuel status	3.59	Empty	1.00
Proximity to obstacles	132.11	Near	0.38
TCA's conflict level	2.25	Resolution advisory	0.71

Output name	Value	Label	Degree
Thrust command	100.00	Max	1.00
Elevator deflection	-20.00	Down	1.00
Autopilot mode suggestion	0.00	Manual	1.00
Stall warning	1.00	Warning	1.00
Fuel reserve command	0.00	Emergency diversion	1.00
Stress reduction command	1.00	Override manual	1.00

Explanation
- Low airspeed and pitch attitude indicate high stall risk; issuing stall advisory is critical. - Engine health degraded; thrust adjusted to ensure performance under failure conditions. - High TCAS conflict level; autopilot mode and manual override may be prioritized for safety.

Main Contributing Factors		
Altitude	756.39	Low
Fuel status	3.59	Empty

FAIL SAFE MODE ACTIVATED

A critical condition was detected
Emergency control outputs were issued.
Fuzzy logic system was overridden for safety response.